LRS31 Asset Pricing and Portfolio Theory

Home Assignment 2

Use RStudio to complete all numerical calculations, data processing, statistical tests, portfolio optimisation, and plots in R. Show analytical derivations where requested. Submit your R script (`.R`) or R Markdown source together with your written answers and relevant R output. Your code must reproduce your reported results.

Assume that you are a portfolio manager considering only the five stocks in the `HA2data_2022` file. It is the end of 2022, and you are planning investments for 2023. Use sample variance and volatility, with $T-1$ in the variance denominator. Use all available historical observations, excluding any observations after 2022. Do not use 2023 observations unless explicitly requested. Use simple returns throughout, expressed as decimals in R and converted to percentages for reporting.

1. Use R to calculate descriptive statistics for the five stocks' simple daily returns. Test each series for normality with the Jarque–Bera test. For how many series can you reject the null hypothesis of normality?

a. Fewer than 2
b. 2
c. 3
d. More than 3

2. Which stock had the lowest average daily bid–ask spread (%) during the sample?

a. Apple
b. Microsoft
c. Tesla
d. JP Morgan or Walmart

3. Use monthly simple returns for this and all remaining questions. If you invest 100% in the stock with the highest average return, what is your average monthly return? What is the average monthly return if you instead select the worst-performing stock? **Answer:** What is the difference between the best and worst average rates of return per annum?

a. Less than 5 percentage points
b. 5–10 percentage points
c. 10–15 percentage points
d. More than 15 percentage points

4. Assume that you can predict each month's best-performing stock and invest 100% in it each month. What is your rate of return per annum? If you instead select the worst-performing stock every month, what is your average monthly return? **Answer:** What is the difference between the best and worst rates of return per annum?

a. Less than 5 percentage points
b. 5–10 percentage points
c. 10–15 percentage points

d. More than 15 percentage points

5. Calculate the simple return on an equally weighted portfolio of all five stocks. Keep the portfolio equally weighted. What is its mean return per annum?

a. Less than 5%
b. 5–10%
c. 10–15%
d. More than 15%

6. Calculate the simple return on a value-weighted portfolio using market capitalisation weights. What is its mean return per annum?

a. Less than 5%
b. 5–10%
c. 10–15%
d. More than 15%

7. Invest the same amount in each stock at the beginning of the period and then make no further trades. Calculate the portfolio's simple returns. What is its mean return per annum? *Hint:* this is the hardest calculation in Q5–Q7.

a. Less than 5%
b. 5–10%
c. 10–15%
d. More than 15%

8. A buy-and-hold portfolio involves investing at the beginning of the period and making no subsequent trades, except possibly reinvesting dividends. Which of the portfolios in Q5 and Q6 most closely resembles a buy-and-hold portfolio?

a. Equally weighted portfolio
b. Value-weighted portfolio

9. Use monthly data to estimate the mean-return vector $\mathbf{R}$ and variance–covariance matrix $\mathbf{\Sigma}$ for all five assets. Both are monthly quantities. You may consult the `Efficient frontier II` sheet of `Chapter_7.xlsx` as a reference, but implement the analysis in R for five assets.

Calculate the risky-asset weights on the capital allocation line (CAL) using

$$\mathbf{w}_{\text{CAL}}(\mu_p) = \mathbf{\Sigma}^{-1}(\mathbf{R} - R_f\mathbf{1})\frac{\mu_p - R_f}{K^2}, \qquad K^2 = A - 2BR_f + CR_f^2,$$

where $\mathbf{1}$ is a vector of ones and

$$A = \mathbf{R}^\top\mathbf{\Sigma}^{-1}\mathbf{R}, \quad B = \mathbf{1}^\top\mathbf{\Sigma}^{-1}\mathbf{R}, \quad C = \mathbf{1}^\top\mathbf{\Sigma}^{-1}\mathbf{1}.$$

Assume a risk-free rate of 0.2% per month. For target returns from 0% to 2% in steps of 0.1% per month, use the weights to calculate portfolio volatility from

$$\text{Var}(R_p) = \mathbf{w}^\top\mathbf{\Sigma}\mathbf{w}.$$

Compare this with the direct formula

$$\sigma^2(\mu_p) = \frac{(\mu_p - R_f)^2}{K^2}.$$

Answer: What is the sum of the stock weights when the target return is 1% per month? *R hint:* use `solve()` for linear systems and `%*%` for matrix multiplication. The risky-asset weights form a 5×1 vector; the remaining weight is invested in the risk-free asset.

a. SUM < 0.2
b. $0.2 <$ SUM < 0.4
c. $0.4 <$ SUM < 0.8
d. $0.8 <$ SUM

10. Plot the minimum-variance frontier of the five risky assets in R, allowing short selling. What is the volatility of the minimum-variance portfolio for a target return of 0.30% per month?

a. Less than 5%
b. 5–10%
c. 10–15%
d. More than 15%